

QNH Budget Planning and Approval Module

Complete Workflow, Database Design, and Backend Behavior

Scope: Financial Year **OPEN** through Financial Year **PRE_CLOSING**

Audience: Database developers, backend developers, frontend developers, reviewers, and future maintainers

Status: Updated target design based on the agreed new workflow

Database: Microsoft SQL Server

Application stack: Node.js / Express / SQL Server / React

1. Purpose of This Document

This document is the main technical reference for redesigning the QNH Budget Planning and Approval module.

It defines:

- the complete workflow from opening a financial year until moving it to **PRE_CLOSING**;
- the purpose and exact relationship of every table in this module;
- when each record is created;
- what the backend validates before each action;
- what the backend inserts or updates after each action;
- which operations must execute in one SQL transaction;
- which statuses are stored and which values are calculated;
- how HOD users enter and resubmit department budgets;
- how Category Budget Managers review department items;
- how hospital-wide category packages are created;
- when package-item records are created;
- how reusable sub-item master records work;
- how the automatic **General** sub-item works;
- where quantity, unit price, specifications, attachments, and future PO links belong;
- how the CFO reviews or returns a complete category package;
- how a Category Budget Manager returns a specific department budget when HOD input is required;
- how change requests work while the year is still **OPEN**;
- why **PRE_CLOSING** is the final approval of the complete annual budget;
- how workflow history, concurrency, validation, and reporting should work.

This document replaces the earlier proposal that used:

- a separate financial-year budget approval table;
 - unnecessary intermediate statuses such as **SUBMITTED** and **UNDER_CATEGORY_REVIEW**;
 - attachments directly on package parent items;
 - no reusable sub-item catalog.
-

2. Final Agreed Business Flow

The new workflow is:

```
Department User / HOD
→ Category Budget Manager
→ CFO
→ PRE_CLOSING
→ Execution
```

The three budget responsibility categories are:

```
IT
Biomedical
General
```

Each department receives:

```
One Department Annual Budget
├─ Three Department Category Budgets
│   ├── IT
│   ├── Biomedical
│   └─ General
```

Each Category Budget Manager prepares one hospital-wide package:

```
Financial Year 2027
├─ Hospital IT Package
├─ Hospital Biomedical Package
└─ Hospital General Package
```

The CFO reviews or returns each complete category package.

The CFO does **not** directly return a department budget. The CFO returns the complete category package to its Category Budget Manager. The Category Budget Manager decides whether:

1. the package can be corrected directly; or
2. a specific department category budget must be returned to its HOD.

The return path is:

```
CFO
→ Category Budget Manager
```

- HOD, only when department input is required
- Category Budget Manager
- CFO

When all three category packages are ready, the CFO moves the financial year:

- OPEN
- PRE_CLOSING

That transition is the CFO's final approval of the complete annual hospital budget.

There is no separate whole-year approval table.

3. Module Boundary

3.1 Included

This module includes:

- Financial Year OPEN
 - Department Budget Initialization
 - Department Budget Preparation
 - Department Category Submission
 - Category Manager Review
 - Category Submission Cutoff
 - Hospital Category Package Preparation
 - Reusable and Year-Specific Sub-Items
 - Sub-Item Pricing and Attachments
 - CFO Category Package Review
 - Controlled Changes While Year Is OPEN
- Financial Year PRE_CLOSING

3.2 Outside This Module

The following activities begin after **PRE_CLOSING** and will be designed separately:

- category budget transfers;
- purchase order linking;
- budget consumption;
- remaining balances;
- execution reporting;
- final financial-year closing.

However, this module defines the sub-item record that the future PO-link module must reference.

4. Roles and Responsibilities

4.1 HOD / Department Budget Manager

The HOD:

- works only in the assigned department;
- sees one complete department budget for the current open financial year;
- sees three category tabs: IT, Biomedical, and General;
- selects generic catalog items;
- enters requested quantities;
- selects a distribution method;
- does not enter unit price;
- does not select a vendor;
- does not define models or detailed technical specifications;
- submits each category independently;
- edits only department items marked **NEEDS_MODIFICATION**;
- cannot modify accepted items directly;
- uses change requests for controlled changes after CFO package completion and before **PRE_CLOSING**.

4.2 Department User

A Department User can assist inside the assigned department according to effective permissions.

The scope comes from:

```
BS_budget_user_roles.department_id
```

The allowed actions come from:

```
Role permissions
+ user permission grants
- user permission denials
```

4.3 Category Budget Manager

A Category Budget Manager:

- is assigned to one budget category;
- sees the assigned category across all departments;
- reviews department items;
- records category-approved quantities;
- accepts items or marks them **NEEDS_MODIFICATION**;
- returns a department category budget when department correction is required;
- sees consolidated hospital demand;

- manages reusable sub-item definitions;
- manages year-specific package sub-items;
- enters package sub-item quantities and unit prices;
- adds specifications and attachments to package sub-items;
- reconciles sub-item quantities with accepted department quantities;
- closes and reopens the category submission window;
- submits the complete hospital-wide category package to CFO;
- handles CFO returns;
- decides whether a CFO return can be fixed at package level or requires HOD input.

The category scope comes from:

```
BS_budget_user_roles.budget_category_id
```

4.4 CFO / Budget Approver

The CFO:

- opens a financial year;
- reviews complete hospital-wide category packages;
- accepts package items or marks them **NEEDS_MODIFICATION**;
- returns a complete category package to the Category Budget Manager;
- may reopen a completed package while the financial year remains **OPEN**;
- does not directly modify department quantities;
- does not directly modify sub-item quantities, prices, or specifications;
- moves the financial year to **PRE_CLOSING** when all conditions are satisfied.

Moving the year to **PRE_CLOSING** is the final approval of the complete annual budget.

4.5 Budget System Administrator

The administrator manages:

- departments;
- budget categories;
- generic catalog items;
- reusable sub-item masters;
- units of measure;
- roles;
- permissions;
- role assignments;
- user permission overrides.

5. Core Design Decisions

5.1 Keep `BS_financial_years`

`BS_financial_years` remains the parent of all year-specific workflow data.

It is required because department budgets, category windows, category packages, change requests, transfers, and PO links must belong to a financial year.

5.2 Remove the Separate Annual Approval Table

Do not create:

```
BS_financial_year_budget_approvals
BS_financial_year_budget_approval_packages
```

Reason:

- while the year is `OPEN`, a completed category package may still be returned;
- a separate approval record would become stale when a package is reopened;
- it would require approval rounds and invalidation logic;
- `pre_closed_by` and `pre_closed_at` already record the CFO and time of final approval.

Final meaning:

```
CFO_REVIEW_COMPLETED
= one category package is accepted but still reopenable while the year is OPEN

PRE_CLOSING
= the complete annual hospital budget is finally approved and planning is locked
```

5.3 One Parent Department Budget per Department and Year

There must be exactly one record for:

```
Financial Year + Department
```

Example:

```
Laboratory Department Budget – 2027
```

This parent provides one clear ID for loading the department's complete annual budget.

5.4 Three Category Budgets under Every Department Budget

Each parent receives exactly three child rows:

```
IT
Biomedical
General
```

The rows are created automatically when the year opens.

5.5 Category Budgets Move Independently

Example:

```
Laboratory 2027

IT          = CATEGORY_REVIEW_COMPLETED
Biomedical  = RETURNED_TO_DEPARTMENT
General     = DRAFT
```

Therefore, the parent department budget does not store a workflow status. Its overall status is calculated from the three children.

5.6 Minimal Stored Statuses

A stored status should exist only when it changes:

- responsibility;
- editability;
- permissions;
- next allowed actions.

Do not store separate statuses only to show that a user opened a page or started reading a record.

5.7 Package Items Are Workflow Entities, Not Stored Totals

`BS_category_budget_package_items` exists because it provides a stable parent for:

- year-specific sub-items;
- CFO item decisions;
- reconciliation status;
- review notes;
- workflow history.

Hospital totals are calculated from department items and are not duplicated in the package-item row.

5.8 Package Items Are Created Lazily

A package-item row is created only when the first HOD submits that catalog item.

Example:

HOD adds Laptop as draft
→ no package item

First department submits Laptop
→ create one Laptop package item

Second department submits Laptop
→ reuse the same package item

5.9 Every Catalog Item Has a Reusable General Sub-Item

When a generic catalog item is created, the backend also creates one reusable default sub-item:

General

When the corresponding package item is created for a year, the backend automatically creates one year-specific **General** package sub-item.

This guarantees that the Category Budget Manager can prepare and price the package even when no detailed model split is needed.

5.10 Price, Attachments, and Future PO Links Belong to Package Sub-Items

The execution hierarchy is:

Generic Catalog Item
→ Reusable Sub-Item Master
→ Year-Specific Package Sub-Item
→ Quantity
→ Unit Price
→ Attachments
→ Future PO Linking

Do not attach these directly to the generic package parent item.

5.11 Selective Editing after Returns

Returning a header does not unlock all children.

Only rows marked:

NEEDS_MODIFICATION

become editable.

Accepted rows remain locked.

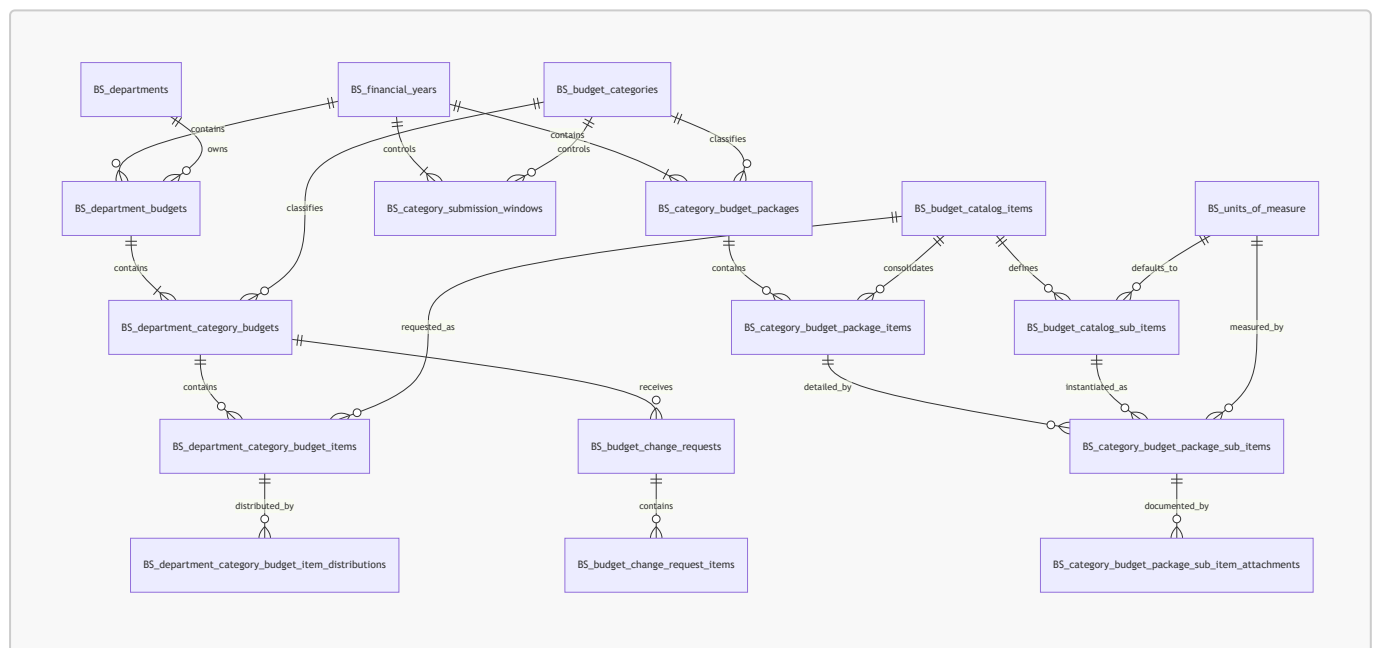
5.12 Multi-Table Commands Are Transactional

Any backend action that changes several related tables must use one SQL transaction.

Examples:

- open a financial year;
- submit a department category budget;
- create a package item and its General sub-item;
- return a department category budget;
- close or reopen a category window;
- submit or return a package;
- apply a change request;
- move the year to **PRE_CLOSING**.

6. High-Level Entity Model



7. Tables Used by This Module

7.1 Access-Control Tables Already Approved

```
BS_budget_roles
BS_budget_permissions
BS_budget_role_permissions
```

```
BS_budget_user_roles  
BS_budget_user_permission_overrides
```

7.2 Master-Data Tables

```
BS_departments  
BS_budget_categories  
BS_units_of_measure  
BS_budget_catalog_items  
BS_budget_catalog_sub_items
```

7.3 Planning and Approval Workflow Tables

```
BS_financial_years  
BS_department_budgets  
BS_department_category_budgets  
BS_department_category_budget_items  
BS_department_category_budget_item_distributions  
BS_category_submission_windows  
BS_category_budget_packages  
BS_category_budget_package_items  
BS_category_budget_package_sub_items  
BS_category_budget_package_sub_item_attachments  
BS_budget_change_requests  
BS_budget_change_request_items  
BS_budget_workflow_history
```

7.4 Calculated Views

Recommended views:

```
VW_category_budget_consolidated_items  
VW_category_budget_department_contributions  
VW_category_package_financial_summary  
VW_department_budget_overall_status  
VW_financial_year_pre_closing_readiness
```

There is no physical package-contributions table because department contributions can be calculated from source records.

8. Final Stored Status Models

8.1 Financial Year

OPEN
PRE_CLOSING
CLOSED

8.2 Department Category Budget Header

DRAFT
IN_CATEGORY_REVIEW
RETURNED_TO_DEPARTMENT
CATEGORY_REVIEW_COMPLETED

Flow:

DRAFT
→ HOD submits
→ IN_CATEGORY_REVIEW

IN_CATEGORY_REVIEW
→ RETURNED_TO_DEPARTMENT
→ HOD resubmits
→ IN_CATEGORY_REVIEW

IN_CATEGORY_REVIEW
→ CATEGORY_REVIEW_COMPLETED

There is no separate **SUBMITTED** status.

Submission metadata is stored in:

submitted_by
submitted_at

8.3 Department Category Budget Item

DRAFT
PENDING_CATEGORY_REVIEW
CATEGORY_ACCEPTED
NEEDS_MODIFICATION

8.4 Category Submission Window

OPEN
CLOSED

8.5 Category Package Header

DRAFT
IN_CFO_REVIEW
RETURNED_BY_CFO
CFO_REVIEW_COMPLETED

Flow:

DRAFT
→ Category Manager submits
→ IN_CFO_REVIEW

IN_CFO_REVIEW
→ RETURNED_BY_CFO
→ Category Manager resubmits
→ IN_CFO_REVIEW

IN_CFO_REVIEW
→ CFO_REVIEW_COMPLETED

CFO_REVIEW_COMPLETED
→ RETURNED_BY_CFO

The last transition is allowed only while the financial year is **OPEN**.

8.6 Package Item CFO Review

Before CFO submission:

`cfo_review_status = NULL`

During CFO review:

PENDING_CFO_REVIEW
CFO_ACCEPTED
NEEDS_MODIFICATION

8.7 Derived Display States

The following values are calculated and are not stored as header statuses.

Department Category: No Requests

```
header.status = DRAFT  
active item count = 0  
category window = CLOSED
```

Display:

```
NO_REQUESTS
```

Department Category: Missed Cutoff

```
header.status = DRAFT  
active item count > 0  
submitted_at IS NULL  
category window = CLOSED
```

Display:

```
NOT_SUBMITTED_AT_CUTOFF
```

Hospital Category: No Requests

```
category window = CLOSED  
active package item count = 0  
no submitted department items exist
```

Display:

```
NO_REQUESTS
```

The package header can remain **DRAFT**.

9. Detailed Table Designs

9.1 **BS_financial_years**

Purpose:

Represents one annual budget cycle and its operational lifecycle.

Column	Type	Null	Notes
id	INT IDENTITY(1,1)	No	Primary key
year	INT	No	Unique calendar/financial year
status	VARCHAR(20)	No	OPEN, PRE_CLOSING, CLOSED
opened_by	INT	No	CFO who opened the year
opened_at	DATETIME2(3)	No	Actual opening time
pre_closed_by	INT	Yes	CFO giving final annual approval
pre_closed_at	DATETIME2(3)	Yes	Final approval / planning lock time
closed_by	INT	Yes	CFO who finally closed execution
closed_at	DATETIME2(3)	Yes	Final closing time
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Last update
row_version	ROWVERSION	No	Optimistic concurrency

Constraints:

```
PK(id)
UNIQUE(year)
CHECK status IN ('OPEN', 'PRE_CLOSING', 'CLOSED')
CHECK pre-closing and closing timestamps match the current status
```

Important:

PRE_CLOSING is the final whole-year budget approval.

9.2 BS_department_budgets

Purpose:

One complete department annual budget container.

Example:

Laboratory Department Budget – 2027

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
financial_year_id	INT	No	FK to financial year
department_id	INT	No	FK to department
created_by	INT	Yes	CFO/system
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Last update
row_version	ROWVERSION	No	Concurrency

Constraints:

```
PK(id)
FK(financial_year_id) → BS_financial_years.id
FK(department_id) → BS_departments.id
UNIQUE(financial_year_id, department_id)
```

No status is stored. The overall state is derived from its three child category budgets.

9.3 BS_department_category_budgets

Purpose:

One IT, Biomedical, or General budget under a department annual budget.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
department_budget_id	BIGINT	No	Parent department annual budget
budget_category_id	INT	No	IT, Biomedical, or General
status	VARCHAR(40)	No	Header workflow state
submitted_by	INT	Yes	Latest submitter
submitted_at	DATETIME2(3)	Yes	Latest submission
returned_by	INT	Yes	Category Manager
returned_at	DATETIME2(3)	Yes	Latest return
return_reason	NVARCHAR(1000)	Yes	Header return reason

Column	Type	Null	Notes
category_review_completed_by	INT	Yes	Category Manager
category_review_completed_at	DATETIME2(3)	Yes	Completion time
created_by	INT	Yes	Creator/system
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Last update
row_version	ROWVERSION	No	Concurrency

Statuses:

DRAFT
IN_CATEGORY_REVIEW
RETURNED_TO_DEPARTMENT
CATEGORY_REVIEW_COMPLETED

Constraints:

PK(id)
FK(department_budget_id) → BS_department_budgets.id
FK(budget_category_id) → BS_budget_categories.id
UNIQUE(department_budget_id, budget_category_id)

9.4 BS_department_category_budget_items

Purpose:

Stores a generic item requested by one department in one category.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
department_category_budget_id	BIGINT	No	Parent category budget
catalog_item_id	INT	No	Generic catalog item
requested_quantity	DECIMAL(18,4)	No	HOD-entered quantity
category_approved_quantity	DECIMAL(18,4)	Yes	Category Manager decision
distribution_method	VARCHAR(40)	No	Annual, quarterly, monthly, custom

Column	Type	Null	Notes
review_status	VARCHAR(40)	No	Item review state
review_note	NVARCHAR(1000)	Yes	Category Manager note
reviewed_by	INT	Yes	Category Manager
reviewed_at	DATETIME2(3)	Yes	Review time
is_active	BIT	No	Soft-delete flag
created_by	INT	Yes	HOD/Department User
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Last update
row_version	ROWVERSION	No	Concurrency

Statuses:

DRAFT
 PENDING_CATEGORY_REVIEW
 CATEGORY_ACCEPTED
 NEEDS_MODIFICATION

Constraints:

PK(id)
 FK(department_category_budget_id) → BS_department_category_budgets.id
 FK(catalog_item_id) → BS_budget_catalog_items.id
 UNIQUE(department_category_budget_id, catalog_item_id)
 CHECK requested_quantity > 0
 CHECK category_approved_quantity IS NULL OR category_approved_quantity >= 0

The database should additionally validate through backend logic that the catalog item's category matches the category budget.

9.5 BS_department_category_budget_item_distributions

Purpose:

Stores the requested quantity distribution across annual, quarterly, monthly, or custom periods.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key

Column	Type	Null	Notes
department_budget_item_id	BIGINT	No	Parent item
period_type	VARCHAR(20)	No	YEAR, QUARTER, MONTH, CUSTOM
period_no	INT	No	Period number
quantity	DECIMAL(18,4)	No	Period quantity
created_at	DATETIME2(3)	No	Creation
updated_at	DATETIME2(3)	Yes	Last update

Constraints:

```
FK(department_budget_item_id) → BS_department_category_budget_items.id
UNIQUE(department_budget_item_id, period_type, period_no)
CHECK quantity >= 0
```

Backend rule:

```
SUM(distribution quantities) = requested_quantity
```

9.6 BS_category_submission_windows

Purpose:

Controls whether departments may submit one category in one financial year.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
financial_year_id	INT	No	Financial year
budget_category_id	INT	No	Category
status	VARCHAR(20)	No	OPEN or CLOSED
closed_by	INT	Yes	Category Manager
closed_at	DATETIME2(3)	Yes	Latest close
close_reason	NVARCHAR(1000)	Yes	Reason
reopened_by	INT	Yes	Category Manager
reopened_at	DATETIME2(3)	Yes	Latest reopen
reopen_reason	NVARCHAR(1000)	Yes	Reason
created_at	DATETIME2(3)	No	Creation

Column	Type	Null	Notes
updated_at	DATETIME2(3)	Yes	Update
row_version	ROWVERSION	No	Concurrency

Constraints:

```
UNIQUE(financial_year_id, budget_category_id)
CHECK status IN ('OPEN', 'CLOSED')
```

9.7 BS_category_budget_packages

Purpose:

One hospital-wide IT, Biomedical, or General package for a financial year.

Example:

```
2027 Hospital IT Package
```

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
financial_year_id	INT	No	Year
budget_category_id	INT	No	Category
status	VARCHAR(40)	No	Package state
submitted_to_cfo_by	INT	Yes	Category Manager
submitted_to_cfo_at	DATETIME2(3)	Yes	Latest submission
returned_by_cfo	INT	Yes	CFO
returned_at	DATETIME2(3)	Yes	Latest return
return_reason	NVARCHAR(1000)	Yes	Package-level reason
cfo_review_completed_by	INT	Yes	CFO
cfo_review_completed_at	DATETIME2(3)	Yes	Completion
created_by	INT	Yes	CFO/system
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Update

Column	Type	Null	Notes
row_version	ROWVERSION	No	Concurrency

Statuses:

DRAFT
IN_CFO_REVIEW
RETURNED_BY_CFO
CFO_REVIEW_COMPLETED

Constraints:

FK(financial_year_id) → BS_financial_years.id
FK(budget_category_id) → BS_budget_categories.id
UNIQUE(financial_year_id, budget_category_id)

9.8 BS_category_budget_package_items

Purpose:

Stable year/category/catalog-item workflow record used to attach package sub-items and CFO decisions.

It does not store duplicated hospital totals.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
category_budget_package_id	BIGINT	No	Package
catalog_item_id	INT	No	Generic item
cfo_review_status	VARCHAR(40)	Yes	Null before CFO submission
cfo_review_note	NVARCHAR(1000)	Yes	CFO note
cfo_reviewed_by	INT	Yes	CFO
cfo_reviewed_at	DATETIME2(3)	Yes	Review time
needs_reconciliation	BIT	No	Source quantities/details changed
is_active	BIT	No	Soft state
created_by	INT	Yes	System/current submitter
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater

Column	Type	Null	Notes
updated_at	DATETIME2(3)	Yes	Update
row_version	ROWVERSION	No	Concurrency

CFO statuses:

```

NULL
PENDING_CFO_REVIEW
CFO_ACCEPTED
NEEDS_MODIFICATION

```

Constraints:

```

FK(category_budget_package_id) → BS_category_budget_packages.id
FK(catalog_item_id) → BS_budget_catalog_items.id
UNIQUE(category_budget_package_id, catalog_item_id)

```

Initial creation values:

```

cfo_review_status = NULL
needs_reconciliation = 1
is_active = 1

```

9.9 BS_budget_catalog_sub_items

Purpose:

Reusable sub-item/model/specification master that can be reused across financial years.

Example:

Generic Item: Computers & Laptops

Reusable Sub-Items:

- General
- Dell Latitude 5450
- HP EliteBook 840

Column	Type	Null	Notes
id	INT IDENTITY(1,1)	No	Primary key
catalog_item_id	INT	No	Generic parent item

Column	Type	Null	Notes
sub_item_code	VARCHAR(100)	No	Stable technical code
name	NVARCHAR(300)	No	Display name
default_specification	NVARCHAR(2000)	Yes	Reusable template
default_unit_of_measure_id	INT	No	Default unit
is_default_general	BIT	No	Identifies fallback sub-item
is_active	BIT	No	Active master
created_by	INT	Yes	Creator
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Update
row_version	ROWVERSION	No	Concurrency

Constraints:

```
FK(catalog_item_id) → BS_budget_catalog_items.id
FK(default_unit_of_measure_id) → BS_units_of_measure.id
UNIQUE(catalog_item_id, sub_item_code)
UNIQUE(catalog_item_id, name)
```

Recommended filtered unique index:

```
Only one active is_default_general = 1 row per catalog_item_id
```

Do not store year-specific quantity, unit price, attachments, supplier, or PO data in this master.

9.10 BS_category_budget_package_sub_items

Purpose:

Year-specific detailed specification and priced execution record inside one package item.

This is the record targeted by future PO linking.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
category_budget_package_item_id	BIGINT	No	Parent package item
catalog_sub_item_id	INT	No	Reusable master

Column	Type	Null	Notes
name	NVARCHAR(300)	No	Year-specific snapshot
specification	NVARCHAR(2000)	Yes	Year-specific snapshot
unit_of_measure_id	INT	No	Year-specific unit
quantity	DECIMAL(18,4)	Yes	Null until prepared
unit_price	DECIMAL(18,6)	Yes	Null until prepared
note	NVARCHAR(1000)	Yes	Manager note
is_active	BIT	No	Soft delete
created_by	INT	Yes	Manager/system
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Last updater
updated_at	DATETIME2(3)	Yes	Update
row_version	ROWVERSION	No	Concurrency

Constraints:

```

FK(category_budget_package_item_id) → BS_category_budget_package_items.id
FK(catalog_sub_item_id) → BS_budget_catalog_sub_items.id
FK(unit_of_measure_id) → BS_units_of_measure.id
UNIQUE(category_budget_package_item_id, catalog_sub_item_id)
CHECK quantity IS NULL OR quantity >= 0
CHECK unit_price IS NULL OR unit_price >= 0

```

The copied `name`, `specification`, and `unit_of_measure_id` preserve the exact year-specific values even if the reusable master changes later.

Calculated amount:

```
quantity × unit_price
```

9.11 BS_category_budget_package_sub_item_attachments

Purpose:

Stores quotations, specifications, comparisons, and other documents for one year-specific package sub-item.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
category_budget_package_sub_item_id	BIGINT	No	Parent sub-item
document_type	VARCHAR(50)	Yes	Quotation, specification, comparison, etc.
original_file_name	NVARCHAR(300)	No	Original file name
storage_key	NVARCHAR(500)	No	Storage reference
mime_type	NVARCHAR(150)	Yes	MIME type
file_size_bytes	BIGINT	No	File size
description	NVARCHAR(500)	Yes	Description
is_active	BIT	No	Active attachment
uploaded_by	INT	No	Uploader
uploaded_at	DATETIME2(3)	No	Upload time
disabled_by	INT	Yes	Removal user
disabled_at	DATETIME2(3)	Yes	Removal time
disabled_reason	NVARCHAR(500)	Yes	Reason

Constraint:

```
FK(category_budget_package_sub_item_id)
→ BS_category_budget_package_sub_items.id
```

If one document applies to the entire parent item, attach it to the **General** package sub-item.

9.12 BS_budget_change_requests

Purpose:

Header for a controlled HOD request to change an already completed budget while the year is still **OPEN**.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
department_category_budget_id	BIGINT	No	Affected department category
status	VARCHAR(40)	No	Change-request workflow
reason	NVARCHAR(2000)	No	Overall justification

Column	Type	Null	Notes
submitted_by	INT	Yes	HOD
submitted_at	DATETIME2(3)	Yes	Submission
category_reviewed_by	INT	Yes	Category Manager
category_reviewed_at	DATETIME2(3)	Yes	Category review
category_note	NVARCHAR(1000)	Yes	Category note
cfo_reviewed_by	INT	Yes	CFO
cfo_reviewed_at	DATETIME2(3)	Yes	CFO review
cfo_note	NVARCHAR(1000)	Yes	CFO note
applied_by	INT	Yes	Applier
applied_at	DATETIME2(3)	Yes	Apply time
created_by	INT	No	Creator
created_at	DATETIME2(3)	No	Creation
updated_by	INT	Yes	Updater
updated_at	DATETIME2(3)	Yes	Update
row_version	ROWVERSION	No	Concurrency

Statuses:

DRAFT
IN_CATEGORY_REVIEW
RETURNED_BY_CATEGORY
IN_CFO_REVIEW
RETURNED_BY_CFO
APPROVED
REJECTED
APPLIED
CANCELLED

9.13 BS_budget_change_request_items

Purpose:

Stores each requested modification inside one change request.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
change_request_id	BIGINT	No	Parent

Column	Type	Null	Notes
change_type	VARCHAR(40)	No	Requested action
existing_department_budget_item_id	BIGINT	Yes	Existing item
catalog_item_id	INT	No	Target generic item
current_requested_quantity	DECIMAL(18,4)	Yes	Snapshot
proposed_requested_quantity	DECIMAL(18,4)	No	Proposed value
proposed_distribution_method	VARCHAR(40)	Yes	Proposed method
description	NVARCHAR(1000)	Yes	Item reason
category_review_status	VARCHAR(40)	No	Category decision
category_review_note	NVARCHAR(1000)	Yes	Note
cfo_review_status	VARCHAR(40)	No	CFO decision
cfo_review_note	NVARCHAR(1000)	Yes	Note
created_at	DATETIME2(3)	No	Creation
updated_at	DATETIME2(3)	Yes	Update

Change types:

```
ADD_ITEM
INCREASE_QUANTITY
DECREASE_QUANTITY
MODIFY_ITEM
```

9.14 BS_budget_workflow_history

Purpose:

Append-only business-readable timeline of important workflow actions.

Column	Type	Null	Notes
id	BIGINT IDENTITY(1,1)	No	Primary key
financial_year_id	INT	Yes	Year context
entity_type	VARCHAR(80)	No	Record type
entity_id	BIGINT	No	Record ID
action	VARCHAR(100)	No	Business event
old_status	VARCHAR(50)	Yes	Previous status
new_status	VARCHAR(50)	Yes	New status

Column	Type	Null	Notes
<code>note</code>	<code>NVARCHAR(1000)</code>	Yes	Human-readable explanation
<code>old_values_json</code>	<code>NVARCHAR(MAX)</code>	Yes	Optional previous details
<code>new_values_json</code>	<code>NVARCHAR(MAX)</code>	Yes	Optional new details
<code>user_role_id</code>	<code>INT</code>	Yes	Acting role assignment
<code>acting_workspace</code>	<code>VARCHAR(80)</code>	Yes	Department, category, CFO, etc.
<code>correlation_id</code>	<code>UNIQUEIDENTIFIER</code>	Yes	Groups events from one command
<code>created_by</code>	<code>INT</code>	No	Actor
<code>created_at</code>	<code>DATETIME2(3)</code>	No	Event time

This table should normally allow inserts only.

10. Automatic General Sub-Item Rules

10.1 When a Generic Catalog Item Is Created

Example:

Computers & Laptops

The backend transaction creates:

1. `BS_budget_catalog_items` row;
2. one `BS_budget_catalog_sub_items` row:

```
sub_item_code = GENERAL
name = General
is_default_general = 1
```

If either insert fails, both must roll back.

10.2 When a Package Item Is First Created

Example:

2027 IT Package
→ Computers & Laptops

The backend transaction:

1. inserts the package item;
2. finds the reusable General sub-item;
3. inserts one year-specific General package sub-item;
4. copies:
 - name;
 - default specification;
 - default unit;
5. leaves:
 - quantity = NULL;
 - unit_price = NULL.

10.3 General Sub-Item Usage

No Detailed Model Split

```
General
Quantity = 50
Unit Price = 4,500
```

Partial Detailed Split

```
Dell Latitude 5450 = 30
HP EliteBook 840 = 15
General = 5
```

Fully Detailed Split

```
Dell Latitude 5450 = 30
HP EliteBook 840 = 20
General = 0
```

The General sub-item remains available for shared documents.

11. Exact Backend and Database Workflow

11.1 CFO Opens a Financial Year

User Action

CFO creates Financial Year 2027.

Backend Validations

1. User has financial-year lifecycle permission.
2. No year exists with `year = 2027`.
3. No other year is `OPEN` or `PRE_CLOSING`.
4. IT, Biomedical, and General categories exist and are active.
5. Active departments exist.
6. Every active generic catalog item has one active General reusable sub-item.

One Database Transaction

A. Insert Financial Year

```
year = 2027
status = OPEN
opened_by = CFO
opened_at = current UTC time
```

B. Create Department Parent Budgets

For each active department:

```
INSERT BS_department_budgets
```

With 30 departments:

```
30 parent rows
```

C. Create Three Category Budgets under Every Parent

For each department parent:

```
IT
Biomedical
General
```

With 30 departments:

```
90 child rows
status = DRAFT
```

D. Create Three Submission Windows

```
2027 + IT + OPEN
2027 + Biomedical + OPEN
2027 + General + OPEN
```

E. Create Three Package Headers

```
2027 + IT + DRAFT
2027 + Biomedical + DRAFT
2027 + General + DRAFT
```

No package-item rows are created yet.

F. Insert Workflow History and Notification Outbox Rows

Example events:

```
FINANCIAL_YEAR_OPENED
DEPARTMENT_BUDGETS_INITIALIZED
CATEGORY_WINDOWS_INITIALIZED
CATEGORY_PACKAGES_INITIALIZED
```

Commit or Rollback

The year must never be partially initialized.

11.2 HOD Loads the Complete Department Budget

The backend resolves:

```
active user-role assignment
→ department_id
→ open financial year
→ BS_department_budgets
→ three BS_department_category_budgets
→ items and distributions
```

Example API result:

```
{
  "departmentBudgetId": 517,
```

```
"financialYear": 2027,
"department": "Laboratory",
"overallStatus": "IN_PROGRESS",
"categories": [
  {
    "category": "IT",
    "status": "DRAFT",
    "derivedDisplayState": null,
    "items": []
  },
  {
    "category": "Biomedical",
    "status": "DRAFT",
    "derivedDisplayState": null,
    "items": []
  },
  {
    "category": "General",
    "status": "DRAFT",
    "derivedDisplayState": null,
    "items": []
  }
]
```

`overallStatus` is calculated.

11.3 HOD Adds a Draft Item

Example:

```
Category: IT
Item: Computers & Laptops
Requested Quantity: 10
Distribution: Quarterly
```

Validations

- financial year is `OPEN`;
- IT window is `OPEN`;
- user is assigned to the department;
- category header is `DRAFT`;
- catalog item belongs to IT;
- catalog item is active;
- no duplicate active item exists;
- quantity is positive;
- distribution is valid.

Transaction

1. insert department category budget item with `review_status = DRAFT`;
2. insert distribution rows;
3. verify distribution sum;
4. insert workflow history.

Package Effect

None.

Draft items do not create package items.

11.4 HOD Updates or Deletes a Draft Item

Editable Conditions

```
header = DRAFT
item = DRAFT
```

or:

```
header = RETURNED_TO_DEPARTMENT
item = NEEDS_MODIFICATION
```

Deletion

Before first submission:

- hard delete may be allowed.

After first submission:

- set `is_active = 0`;
 - preserve history;
 - mark the related package item `needs_reconciliation = 1`.
-

11.5 HOD Submits a Department Category Budget

Example:

```
Laboratory submits IT budget
```

Validations

- year is **OPEN**;
- category window is **OPEN**;
- header is **DRAFT** or **RETURNED_TO_DEPARTMENT**;
- at least one active item exists;
- every editable item is complete;
- quantities are valid;
- distributions reconcile;
- user belongs to the department.

One Transaction

A. Update Item Review States

First submission:

```
DRAFT → PENDING_CATEGORY_REVIEW
```

Resubmission:

```
NEEDS_MODIFICATION → PENDING_CATEGORY_REVIEW
```

Previously accepted items remain:

```
CATEGORY_ACCEPTED
```

B. Update Header

```
status = IN_CATEGORY_REVIEW  
submitted_by = current user  
submitted_at = now
```

There is no intermediate **SUBMITTED** status.

C. Ensure Package Items Exist

For every distinct submitted catalog item:

1. locate the package by financial year and category;
2. locate the package item by package and catalog item;
3. if missing, create it;
4. if existing, reuse it;
5. set **needs_reconciliation = 1**;

6. set `is_active = 1`.

D. Create General Package Sub-Item When Package Item Is New

If the package item was inserted:

1. find the active reusable General sub-item;
2. insert one year-specific package sub-item;
3. copy its name, specification, and unit;
4. leave quantity and unit price null.

E. History and Notifications

Insert:

```
DEPARTMENT_CATEGORY_SUBMITTED  
PACKAGE_ITEM_CREATED, when applicable  
GENERAL_PACKAGE_SUB_ITEM_CREATED, when applicable
```

Queue Category Manager notifications.

11.6 Two Departments Submit the Same Generic Item

Example:

```
Laboratory submits Laptop  
Radiology submits Laptop
```

The second submission:

- does not create a second package item;
- does not create a second General package sub-item;
- reuses the existing package item;
- sets `needs_reconciliation = 1`.

The unique constraint protects concurrent submissions.

11.7 Category Manager Reviews a Department Item

The header is already:

```
IN_CATEGORY_REVIEW
```

No separate "start review" transition is needed.

Accept

Example:

```
Requested = 10  
Category Approved = 8
```

Update:

```
review_status = CATEGORY_ACCEPTED  
category_approved_quantity = 8  
reviewed_by = manager  
reviewed_at = now
```

Needs Modification

Update:

```
review_status = NEEDS_MODIFICATION  
review_note = required  
reviewed_by = manager  
reviewed_at = now
```

The related package item remains:

```
needs_reconciliation = 1
```

11.8 Category Manager Returns the Department Category Budget

Validations

- header is **IN_CATEGORY_REVIEW**;
- at least one item is **NEEDS_MODIFICATION**;
- every active item has a decision;
- no item remains **PENDING_CATEGORY_REVIEW**.

Transaction

```
header:  
IN_CATEGORY_REVIEW → RETURNED_TO_DEPARTMENT
```

Update return metadata.

Accepted items remain locked.

Only **NEEDS_MODIFICATION** items are editable.

Notify HOD.

11.9 HOD Corrects and Resubmits

The backend allows edits only where:

```
header = RETURNED_TO_DEPARTMENT
item = NEEDS_MODIFICATION
```

After changes:

```
related package item.needs_reconciliation = 1
```

On resubmission:

```
header:
RETURNED_TO_DEPARTMENT → IN_CATEGORY_REVIEW

items:
NEEDS_MODIFICATION → PENDING_CATEGORY_REVIEW
```

Accepted items remain unchanged.

11.10 Category Manager Completes Department Review

Allowed only when every active item is:

```
CATEGORY_ACCEPTED
```

Update:

```
header.status = CATEGORY_REVIEW_COMPLETED
category_review_completed_by = manager
category_review_completed_at = now
```

11.11 Consolidated Category View

The Category Manager sees:

Department View

```
Laboratory
- Laptop: requested 10, approved 8

Radiology
- Laptop: requested 15, approved 15
```

Consolidated View

```
Laptop
Total Requested = 25
Total Category Approved = 23
```

Totals are calculated through joins and `SUM()`.

No physical contribution table is required.

For final package readiness, the approved total includes active department items with:

```
review_status = CATEGORY_ACCEPTED
```

11.12 Category Manager Uses Reusable Sub-Items

For package item:

```
Computers & Laptops
```

The Category Manager can:

- keep `General`;
- select an existing reusable sub-item;
- create a new reusable sub-item and immediately instantiate it for the year;
- deactivate a year-specific package sub-item;
- update the year-specific specification without changing the reusable master.

Example:

```
Dell Latitude 5450 = 15  
HP EliteBook 840 = 10  
General = 2
```

11.13 Package Reconciliation

The backend calculates:

```
Accepted Department Total  
=  
SUM(category_approved_quantity for accepted active department items)
```

It also calculates:

```
Package Sub-Item Total  
=  
SUM(quantity for active package sub-items)
```

The package item is reconciled only when:

```
Package Sub-Item Total = Accepted Department Total
```

and every active sub-item with quantity greater than zero has:

- unit of measure;
- unit price;
- valid name;
- valid specification when required.

Then:

```
needs_reconciliation = 0
```

Otherwise:

```
needs_reconciliation = 1
```

11.14 Category Manager Uploads Attachments

Attachments must reference:

```
BS_category_budget_package_sub_items.id
```

Examples:

- Dell quotation → Dell sub-item;
- HP specification → HP sub-item;
- parent-wide document → General sub-item.

A previously submitted attachment should be disabled rather than hard-deleted.

11.15 Category Manager Closes a Submission Window

Purpose:

Stop new department submissions for one category so the package population becomes stable.

Validations

- year is **OPEN**;
- window is **OPEN**;
- user owns the category;
- close reason is provided when required.

Effects

```
window.status = CLOSED
```

Do not rewrite department category header statuses.

Derived states are calculated:

```
DRAFT + zero items  
→ display NO_REQUESTS
```

```
DRAFT + active items + never submitted  
→ display NOT_SUBMITTED_AT_CUTOFF
```

Already submitted/reviewed budgets continue in the workflow.

Closing IT does not close Biomedical or General.

11.16 Reopen a Submission Window

Allowed only while the year is **OPEN**.

Required:

```
reopen_reason
```

The window may be reopened when:

- late department submission is accepted;
- Category Manager needs department input after CFO return;
- a controlled change requires department correction.

Any changed source quantity sets:

```
package_item.needs_reconciliation = 1
```

11.17 Category Manager Submits the Package to CFO

Validations

- financial year is **OPEN**;
- category window is **CLOSED**;
- package is **DRAFT** or **RETURNED_BY_CFO**;
- at least one active package item exists, unless derived no-requests applies;
- no included department item is pending or needs modification;
- every active package item has **needs_reconciliation = 0**;
- active package sub-item quantities reconcile;
- prices are present;
- required attachments exist;
- there are no unresolved department returns for included records.

Transaction

Header

```
DRAFT or RETURNED_BY_CFO  
→ IN_CFO_REVIEW
```

Update submission metadata.

Package Items

First submission:

NULL → PENDING_CFO_REVIEW

Resubmission:

NEEDS_MODIFICATION → PENDING_CFO_REVIEW

Previously accepted items remain:

CFO_ACCEPTED

Record history and notify CFO.

11.18 CFO Reviews Package Items

Accept

PENDING_CFO_REVIEW → CFO_ACCEPTED

Store reviewer and time.

Needs Modification

PENDING_CFO_REVIEW → NEEDS_MODIFICATION

A review note is required.

The CFO cannot directly edit package content.

11.19 CFO Returns the Complete Category Package

Validations

- package is **IN_CFO_REVIEW** or **CFO_REVIEW_COMPLETED**;
- at least one item is **NEEDS_MODIFICATION**;
- no active package item remains **PENDING_CFO_REVIEW**.

Transaction

```
package.status = RETURNED_BY_CFO
```

Accepted package items remain locked.

Only package items marked **NEEDS_MODIFICATION** become editable.

The CFO returns the complete package, not a department budget.

12. CFO Return Cases

12.1 Package-Level Correction Only

Examples:

- wrong unit price;
- missing quotation;
- incorrect technical specification;
- wrong sub-item split;
- attachment missing;
- package sub-item quantity mismatch.

The Category Manager updates only:

- affected package sub-items;
- affected package sub-item attachments;
- affected package item reconciliation.

No department record changes.

After correction:

```
needs_reconciliation = 0
```

Then the package is resubmitted.

12.2 Department Input Is Required

Examples:

- department quantity is unclear;
- wrong generic item was selected;
- business need must change;
- distribution is incorrect.

The Category Manager transaction:

1. keeps package **RETURNED_BY_CFO**;
2. keeps affected package item **NEEDS_MODIFICATION**;
3. sets **needs_reconciliation = 1**;
4. reopens the category window when necessary;
5. marks the affected department item **NEEDS_MODIFICATION**;
6. changes its header to **RETURNED_TO_DEPARTMENT**;
7. records the reason;
8. notifies HOD.

The HOD edits only the affected department item.

After HOD resubmission and Category Manager acceptance:

- totals recalculate;
- package sub-items are reconciled;
- affected package item returns to CFO review.

12.3 No Source Department Item Remains

If no active submitted source remains for a package item:

```
package_item.is_active = 0  
package_item.needs_reconciliation = 1
```

Do not hard-delete package items, sub-items, attachments, or history after they have entered workflow.

If the item becomes relevant again:

```
is_active = 1  
cfo_review_status = NULL  
needs_reconciliation = 1
```

13. Completing and Reopening Category Package Review

13.1 Complete Review

When every active package item is:

```
CFO_ACCEPTED
```

update:

```
package.status = CFO_REVIEW_COMPLETED  
cfo_review_completed_by = CFO  
cfo_review_completed_at = now
```

13.2 Reopen before PRE_CLOSING

While the financial year remains **OPEN**, CFO may return a completed package:

```
CFO_REVIEW_COMPLETED → RETURNED_BY_CFO
```

The CFO must select affected package items and mark them **NEEDS_MODIFICATION**.

This immediately blocks **PRE_CLOSING**.

After **PRE_CLOSING**, this transition is not allowed.

14. Change Request Workflow While Year Is OPEN

14.1 When a Change Request Is Needed

A direct edit is not allowed after an affected package item has been accepted by CFO.

The HOD creates a change request for:

- add item;
- increase quantity;
- decrease quantity;
- modify item;
- modify distribution.

14.2 Eligibility

A change request is allowed only when:

- financial year is **OPEN**;
- affected department category was previously reviewed;
- affected category package is **CFO_REVIEW_COMPLETED**;
- user has department permission;
- no conflicting open request exists.

No separate annual approval record is required.

14.3 Review

The change request moves:

```
DRAFT
→ IN_CATEGORY_REVIEW
→ IN_CFO_REVIEW
→ APPROVED / REJECTED / RETURNED
```

Returned change requests are edited as change-request records, not as approved budget records.

14.4 Apply an Approved Change

One transaction:

1. update or create the affected department item;
2. set its review status to `PENDING_CATEGORY_REVIEW`;
3. set department category header to `IN_CATEGORY_REVIEW`;
4. ensure package item and General package sub-item exist for a new generic item;
5. set affected package item:
 - `needs_reconciliation = 1`;
 - `cfo_review_status = NEEDS_MODIFICATION`;
6. set category package:
 - `CFO_REVIEW_COMPLETED → RETURNED_BY_CFO`;
7. mark change request `APPLIED`;
8. insert workflow history;
9. notify users.

Unaffected department and package items remain accepted and locked.

After the affected flow completes, the package returns to:

```
CFO_REVIEW_COMPLETED
```

There is no annual-approval invalidation table.

15. CFO Moves the Year to PRE_CLOSING

This is the final whole-year budget approval.

15.1 Eligibility

The backend must revalidate all conditions inside the final transaction.

Financial Year

```
status = OPEN
```

Category Windows

All three windows are:

```
CLOSED
```

Category Packages

For each category, either:

```
package.status = CFO_REVIEW_COMPLETED
```

or the category has a derived no-requests condition:

```
window = CLOSED  
no submitted department items  
no active package items  
package remains DRAFT
```

Department Category Budgets

None may be:

```
IN_CATEGORY_REVIEW  
RETURNED_TO_DEPARTMENT
```

A **DRAFT** header is acceptable only when it is a derived:

```
NO_REQUESTS  
or  
NOT_SUBMITTED_AT_CUTOFF
```

Department Items

No active item may be:

```
PENDING_CATEGORY_REVIEW  
NEEDS_MODIFICATION
```

Package Items

No active item may be:

```
PENDING_CFO_REVIEW  
NEEDS_MODIFICATION
```

No active package item may have:

```
needs_reconciliation = 1
```

Change Requests

No open change request may exist.

15.2 Transaction

```
BS_financial_years.status = PRE_CLOSING  
pre_closed_by = CFO  
pre_closed_at = now  
updated_by = CFO  
updated_at = now
```

Insert workflow history:

```
FINANCIAL_YEAR_PRE_CLOSED
```

Queue notifications.

15.3 Result

After commit:

- normal department editing stops;
- category windows cannot reopen;
- category packages cannot be returned;
- new change requests stop;
- transfer and PO execution modules may begin.

16. Future PO-Linking Relationship

The future PO-link table must reference:

```
BS_category_budget_package_sub_items.id
```

Example:

```
2027 IT Package  
→ Computers & Laptops  
→ Dell Latitude 5450  
→ PO line
```

It must not reference only the parent generic package item because:

- approved quantity is sub-item specific;
- unit price is sub-item specific;
- attachments are sub-item specific;
- model/specification is sub-item specific;
- remaining executable quantity is sub-item specific.

17. Calculated Values and Views

17.1 Department Parent Overall Status

Calculated from three category children.

Possible API display values:

```
NOT_STARTED  
IN_PROGRESS  
RETURNED  
PARTIALLY_COMPLETED  
ALL_CATEGORIES_COMPLETED
```

Do not store this on `BS_department_budgets`.

17.2 Consolidated Requested Quantity

```
SUM(requested_quantity)
```

Group by:

```
financial_year  
category
```



```
catalog item
```

17.3 Consolidated Category-Approved Quantity

```
SUM(category_approved_quantity)
```

Include only active accepted department items.

17.4 Department Contributions

Calculated by joining:

```
package
→ package item
→ financial year/category/catalog item
→ department category budget items
→ department budget
→ department
```

17.5 Package Sub-Item Total Quantity

```
SUM(active package sub-item quantity)
```

17.6 Package Item Amount

```
SUM(quantity × unit_price)
```

17.7 Category Package Amount

```
SUM(all active package sub-item amounts)
```

17.8 Derived No-Request and Missed-Cutoff States

Calculated from:

- category window status;
- department category header status;
- active item count;
- submitted timestamp.

18. Backend Architecture

18.1 Controllers

Controllers should:

- parse request input;
- resolve authenticated user;
- resolve active `user_role_id` / workspace;
- call one service;
- return service output;
- avoid workflow business rules.

18.2 Services

Services should:

- validate permission and scope;
- validate current status;
- start transactions;
- call repositories;
- update all affected entities atomically;
- write workflow history;
- write notification outbox rows;
- return domain-specific errors.

18.3 Repositories

Repositories should:

- execute SQL only;
- accept an optional transaction;
- avoid business decisions;
- return records and affected-row counts.

18.4 Recommended Services

```
openFinancialYearService
getDepartmentBudgetService
saveDepartmentBudgetItemService
deleteDepartmentBudgetItemService
submitDepartmentCategoryBudgetService
reviewDepartmentBudgetItemService
returnDepartmentCategoryBudgetService
completeDepartmentCategoryReviewService

closeCategorySubmissionWindowService
reopenCategorySubmissionWindowService
```

```

createCatalogSubItemService
updateCatalogSubItemService
savePackageSubItemService
disablePackageSubItemService
uploadPackageSubItemAttachmentService
disablePackageSubItemAttachmentService
reconcilePackageItemService

submitCategoryPackageToCfoService
reviewPackageItemService
returnCategoryPackageService
completeCategoryPackageReviewService
reopenCompletedCategoryPackageService

createBudgetChangeRequestService
submitBudgetChangeRequestService
reviewBudgetChangeRequestByCategoryService
reviewBudgetChangeRequestByCfoService
applyApprovedBudgetChangeRequestService

moveFinancialYearToPreClosingService

```

Removed services:

```

approveFinancialYearBudgetService
startCategoryReviewService
startCfoPackageReviewService
uploadPackageItemAttachmentService

```

19. Recommended API Actions

```

POST  /api/financial-years
POST  /api/financial-years/:id/pre-close

GET   /api/department-budgets/current
GET   /api/department-budgets/:id

POST  /api/department-category-budgets/:id/items
PATCH /api/department-category-budget-items/:id
DELETE /api/department-category-budget-items/:id
POST  /api/department-category-budgets/:id/submit

GET   /api/category-management/:categoryId/departments
GET   /api/category-management/:categoryId/consolidated-items
POST  /api/department-category-budget-items/:id/review
POST  /api/department-category-budgets/:id/return
POST  /api/department-category-budgets/:id/complete-review

```

```

POST    /api/category-submission-windows/:id/close
POST    /api/category-submission-windows/:id/reopen

GET     /api/budget-catalog-items/:id/sub-items
POST    /api/budget-catalog-items/:id/sub-items
PATCH  /api/budget-catalog-sub-items/:id

GET     /api/category-budget-packages/:id
POST    /api/category-budget-package-items/:id/sub-items
PATCH  /api/category-budget-package-sub-items/:id
DELETE  /api/category-budget-package-sub-items/:id

POST    /api/category-budget-package-sub-items/:id/attachments
DELETE  /api/category-budget-package-sub-item-attachments/:id

POST    /api/category-budget-packages/:id/submit-to-cfo
POST    /api/category-budget-package-items/:id/cfo-review
POST    /api/category-budget-packages/:id/return
POST    /api/category-budget-packages/:id/complete-review
POST    /api/category-budget-packages/:id/reopen

POST    /api/budget-change-requests
POST    /api/budget-change-requests/:id/submit
POST    /api/budget-change-requests/:id/category-review
POST    /api/budget-change-requests/:id/cfo-review
POST    /api/budget-change-requests/:id/apply

```

20. Transaction Boundaries

Action	Main Tables
Create catalog item	catalog item, reusable General sub-item, history
Open financial year	year, department parents, category children, windows, packages, history, outbox
Submit department category	header, items, package items, General package sub-items, history, outbox
Return department category	header, item states, package reconciliation flag, history, outbox
Complete department review	header, items, history
Close/reopen window	window, history, outbox
Create package sub-item	reusable master when needed, package sub-item, package item reconciliation, history
Upload attachment	attachment, history

Action	Main Tables
Submit package to CFO	package, package items, history, outbox
Return package	package, package item states, history, outbox
Reopen completed package	package, affected items, history, outbox
Apply approved change	department item, category header, package, package item, package sub-item creation when required, change request, history, outbox
Move to PRE_CLOSING	financial year, history, outbox

External email and file-transfer work should not keep the SQL transaction open. Store durable queue/storage metadata first, then process externally.

21. Concurrency and Idempotency

21.1 ROWVERSION

Mutable workflow tables should contain:

```
row_version ROWVERSION
```

The frontend sends the version it loaded.

Backend update:

```
WHERE id = @id
AND row_version = @expected_row_version
```

Zero affected rows means a concurrency conflict.

21.2 Critical Commands

Submission, return, completion, change application, and PRE_CLOSING must:

1. read the current row inside the transaction;
2. lock or version-check it;
3. validate the status immediately before update;
4. commit only if all related operations succeed.

21.3 Concurrent First Package-Item Creation

Two departments may submit the same generic item at the same time.

Final protection:

```
UNIQUE(category_budget_package_id, catalog_item_id)
```

Backend behavior:

1. attempt insert;
2. if duplicate key, read existing row;
3. ensure General package sub-item exists;
4. continue idempotently.

21.4 Concurrent General Package Sub-Item Creation

Final protection:

```
UNIQUE(category_budget_package_item_id, catalog_sub_item_id)
```

22. Validation Ownership

22.1 Database Must Enforce

- primary keys;
- foreign keys;
- unique combinations;
- valid stored status values;
- positive/non-negative quantities and prices;
- one department parent per year;
- one category child per parent;
- one package per year/category;
- one package item per package/catalog item;
- one package sub-item per package item/reusable sub-item;
- one active reusable General sub-item per generic catalog item.

22.2 Backend Must Enforce

- permissions;
- acting workspace;
- department/category scope;
- allowed status transitions;
- editable-item rules;
- catalog/category match;
- distribution sum;

- package quantity reconciliation;
- package pricing readiness;
- attachment policy;
- submission-window rules;
- CFO return paths;
- change-request eligibility;
- **PRE_CLOSING** readiness.

22.3 Frontend Should Enforce for UX

- disable invalid controls;
- show only editable items;
- show return reasons;
- show package reconciliation errors;
- send **row_version**;
- display backend readiness failures.

The frontend is not the final security boundary.

23. Important Constraints and Indexes

23.1 Unique Constraints

```
BS_financial_years:
UNIQUE(year)

BS_department_budgets:
UNIQUE(financial_year_id, department_id)

BS_department_category_budgets:
UNIQUE(department_budget_id, budget_category_id)

BS_department_category_budget_items:
UNIQUE(department_category_budget_id, catalog_item_id)

BS_department_category_budget_item_distributions:
UNIQUE(department_budget_item_id, period_type, period_no)

BS_category_submission_windows:
UNIQUE(financial_year_id, budget_category_id)

BS_category_budget_packages:
UNIQUE(financial_year_id, budget_category_id)

BS_category_budget_package_items:
UNIQUE(category_budget_package_id, catalog_item_id)

BS_budget_catalog_sub_items:
```

```
UNIQUE(catalog_item_id, sub_item_code)
UNIQUE(catalog_item_id, name)

BS_category_budget_package_sub_items:
UNIQUE(category_budget_package_item_id, catalog_sub_item_id)
```

23.2 Recommended Filtered Unique Index

One active General reusable sub-item per catalog item

Conceptual filter:

```
is_default_general = 1
AND is_active = 1
```

23.3 Query Indexes

Recommended indexes include:

```
Department category budgets:
(department_budget_id, status)

Department items:
(department_category_budget_id, review_status, is_active)
(catalog_item_id, review_status, is_active)

Packages:
(financial_year_id, budget_category_id, status)

Package items:
(category_budget_package_id, cfo_review_status, is_active)
(catalog_item_id, is_active)

Package sub-items:
(category_budget_package_item_id, is_active)

Workflow history:
(entity_type, entity_id, created_at)
(financial_year_id, created_at)
(created_by, created_at)
```

24. Important Queries and Views

24.1 Load Complete Department Budget

Input:

```
financial_year_id  
department_id
```

Path:

```
BS_department_budgets  
→ BS_department_category_budgets  
→ BS_department_category_budget_items  
→ distributions
```

24.2 Category Manager Consolidated View

Group by catalog item and calculate:

```
total requested  
total category approved  
department contribution count  
pending review count  
needs modification count
```

24.3 Department Contributions View

Recommended output:

```
package_item_id  
department_id  
department_name  
department_category_budget_id  
department_budget_item_id  
requested_quantity  
category_approved_quantity  
review_status
```

24.4 CFO Package View

Return:

- package header;
- package items;
- calculated requested totals;

- calculated category-approved totals;
- department contributions;
- active package sub-items;
- sub-item quantities;
- unit prices;
- calculated amounts;
- sub-item attachments;
- CFO review state;
- workflow history.

24.5 PRE_CLOSING Readiness

Return structured blockers:

```
{
  "eligible": false,
  "reasons": [
    "Biomedical package is returned by CFO",
    "Laboratory General budget is returned to department",
    "One package item needs reconciliation",
    "Two change requests are still open"
  ]
}
```

25. Empty and Special Cases

25.1 Department Has No Request in One Category

The child header remains **DRAFT**.

After window closure, API derives:

```
NO_REQUESTS
```

25.2 Department Has Draft Items but Misses Cutoff

The child header remains **DRAFT**.

After window closure, API derives:

```
NOT_SUBMITTED_AT_CUTOFF
```

Those draft items are excluded from package calculations.

25.3 Entire Hospital Has No Request for a Category

The package header remains **DRAFT**.

After the window is closed and no submitted source/package items exist, API derives:

```
NO_REQUESTS
```

This derived condition is acceptable for **PRE_CLOSING**.

25.4 Catalog Item Becomes Inactive

Historical department, package, and sub-item rows remain.

Inactive master records cannot be selected for new drafts.

25.5 Department Becomes Inactive after Year Initialization

Existing year-specific department budgets remain for history.

25.6 Notification Failure

The business transaction remains committed if a durable notification/outbox row was inserted successfully.

25.7 File Upload Failure

Do not commit attachment metadata until durable storage succeeds, or use a staged upload state and finalize metadata afterward.

25.8 User Has Multiple Roles

Every command must resolve:

```
actor user
active user_role_id
acting workspace
permission
department/category scope
```

26. Implementation Order

Implement and approve one table at a time in this order:

1. **BS_financial_years**
2. **BS_department_budgets**
3. **BS_department_category_budgets**

4. `BS_department_category_budget_items`
 5. `BS_department_category_budget_item_distributions`
 6. `BS_category_submission_windows`
 7. `BS_category_budget_packages`
 8. `BS_category_budget_package_items`
 9. `BS_budget_catalog_sub_items`
 10. `BS_category_budget_package_sub_items`
 11. `BS_category_budget_package_sub_item_attachments`
 12. calculated contribution and summary views
 13. `BS_budget_change_requests`
 14. `BS_budget_change_request_items`
 15. `BS_budget_workflow_history`
 16. backend repositories
 17. backend services
 18. API controllers/routes
 19. frontend workflows
 20. notification outbox and workers
 21. reports
 22. future PO-link module targeting package sub-items
-

27. Acceptance Criteria

27.1 Financial-Year Initialization

- opening a year creates one department parent per active department;
- each parent receives exactly three category budgets;
- exactly three windows are created;
- exactly three package headers are created;
- no package items are created initially;
- the action is atomic.

27.2 Catalog and Sub-Item Masters

- every generic catalog item has exactly one active General reusable sub-item;
- a new catalog item and its General sub-item are created atomically;
- reusable sub-items do not store year-specific prices or quantities.

27.3 HOD Workflow

- HOD sees one annual department budget and three categories;
- category submission is independent;
- draft items do not create package items;
- first submission creates or reuses one package item;
- a new package item creates one General package sub-item;
- accepted department items remain locked;
- only `NEEDS_MODIFICATION` department items are editable.

27.4 Category Manager Workflow

- manager sees only the assigned category;
- department and consolidated views use the same source data;
- approved quantity can differ from requested quantity;
- totals are calculated;
- package sub-item quantities must reconcile;
- pricing and attachments belong to package sub-items;
- package cannot be submitted to CFO while its window is open;
- package cannot be submitted while any active package item needs reconciliation.

27.5 CFO Workflow

- CFO reviews complete category packages;
- CFO can accept or return package items;
- CFO cannot directly edit package content;
- package return unlocks only affected package items;
- CFO returns the package to the Category Manager, never directly to HOD;
- Category Manager returns a department budget only when department input is required;
- CFO can reopen a completed package only while year is **OPEN**.

27.6 Change Requests

- direct edits are blocked after CFO package completion;
- change requests exist only while year is **OPEN**;
- only affected records repeat review;
- unaffected accepted items remain locked;
- applying a change returns the affected package to **RETURNED_BY_CFO**;
- no annual approval table is used.

27.7 PRE_CLOSING

- **PRE_CLOSING** is the final whole-year approval;
- all category windows must be closed;
- every nonempty package must be **CFO_REVIEW_COMPLETED**;
- derived empty packages are allowed;
- no department or package item may need modification;
- no package item may need reconciliation;
- no open change request may exist;
- after pre-close, package returns and normal edits are blocked.

27.8 PO-Link Readiness

- future PO links target **BS_category_budget_package_sub_items.id**;
- quantities, prices, attachments, and specifications are available on the target hierarchy.

28. Glossary

Term	Meaning
Department Budget	Complete annual container for one department and year
Department Category Budget	IT, Biomedical, or General child under the department budget
Department Budget Item	Generic item requested by the department
Category Submission Window	Controls whether departments may submit one category
Category Package	Hospital-wide package for one category and year
Package Item	Stable consolidated generic-item workflow record
Reusable Sub-Item	Master model/specification reusable in different years
General Sub-Item	Automatic fallback sub-item for every generic catalog item
Package Sub-Item	Year-specific priced/specification record inside a package item
Category Accepted	Department item accepted by Category Manager
CFO Accepted	Package item accepted by CFO
Needs Reconciliation	Package sub-items no longer match accepted department quantities/details
PRE_CLOSING	CFO's final whole-year approval and planning lock
Workflow History	Append-only business timeline of important actions

29. Final Workflow Summary

CFO opens financial year

- one parent budget is created for each active department
- three category budgets are created under each parent
- three category windows are opened
- three hospital category package headers are created

HOD enters draft generic items

- no package item is created yet

HOD submits one category

- header becomes IN_CATEGORY_REVIEW
- draft/returned items become PENDING_CATEGORY_REVIEW
- package item is created or reused
- General year-specific package sub-item is created when package item is new

Category Manager reviews department items

- CATEGORY_ACCEPTED
- or
- NEEDS_MODIFICATION

If correction is required

- department category header becomes RETURNED_TO_DEPARTMENT
- HOD edits only affected items
- resubmission returns header to IN_CATEGORY_REVIEW

When all items are accepted

- department category becomes CATEGORY_REVIEW_COMPLETED

Category Manager prepares hospital package

- uses reusable sub-item masters
- keeps or splits General sub-item
- enters year-specific quantities, unit prices, and specifications
- uploads attachments to package sub-items
- reconciles package sub-item quantity with accepted department quantity

Category Manager closes category window

- no new ordinary submissions
- empty/missed-cutoff states are derived, not stored

Category Manager submits package

- package becomes IN_CFO_REVIEW
- relevant package items become PENDING_CFO_REVIEW

CFO reviews package items

- CFO_ACCEPTED
- or
- NEEDS_MODIFICATION

If CFO returns package

- package becomes RETURNED_BY_CFO
- Category Manager fixes package details
- or
- Category Manager returns a specific department category budget to HOD

When every active package item is accepted

- package becomes CFO_REVIEW_COMPLETED

While year remains OPEN

- CFO may reopen a completed package
- controlled change requests may reopen affected records

When all three packages are complete or valid derived no-request categories and all other blockers are resolved

- CFO moves financial year from OPEN to PRE_CLOSING

PRE_CLOSING

- final approval of the complete annual budget
- normal planning changes stop
- transfer and PO execution modules may begin

30. Technical Implementation Basis

The design uses standard SQL Server capabilities:

- explicit transactions for all-or-nothing multi-table operations;
- primary and foreign keys for referential integrity;
- unique and check constraints for data invariants;
- filtered unique indexes for conditional uniqueness;
- aggregate queries and views for calculated totals and contributions;
- **ROWVERSION** for optimistic concurrency;
- short transactions and revalidation for critical workflow transitions.

The backend service layer owns workflow rules. The database provides final protection against structurally invalid data.

31. Technical References

- [BEGIN TRANSACTION \(Transact-SQL\)](#)
- [Primary and foreign key constraints](#)
- [rowversion \(Transact-SQL\)](#)
- [Create filtered indexes](#)
- [Transaction locking and row versioning guide](#)